Quantum Mechanics II — Exercise Sheet 12

Survival, escape and decay

Variant: Questions only

Exercise 1 (A uniform energy band). A normalized state's energy density is $1/(2a)$ on $[E_c - a, E_c + a]$ and zero outside, with $E_c > a > 0$. (i) Find its survival amplitude and probability, the first positive zero, and its short-time expansion. (ii) Compare the expansion with its energy variance.

Exercise 2 (Decay without escape). On $L^2(\mathbb{R})$ let $H = Q$ and $\psi = \mathbf{1}_{[0,1]}$. (i) Compute its survival probability and the probability of remaining in $[0, 1]$ at time t. (ii) Explain why the limit of the survival probability alone does not prove spatial escape.

Exercise 3 (A semibounded model with an algebraic tail). A state's energy density is $w(E) = a^{-1}e^{-E/a}\mathbf{1}_{[0,\infty)}(E)$, where $a > 0$. (i) Compute its survival amplitude, probability, mean energy and variance. (ii) Find the leading long-time decay and decide whether a constant exponential decay rate fits all times.

Exercise 4 (A finite mean with infinite energy variance). A state's energy density is $w(E) = 2a^2/(E + a)^3$ for $E \geq 0$, zero for $E < 0$, where $a > 0$. Check normalization and compute $\mathrm{Prob}(E > R)$ for $R \geq 0$. (i) Find its mean energy and decide whether the second energy moment is finite. Does the quadratic short-time theorem in Lecture 12 apply? (ii) Explain whether the existence of a first energy moment alone determines an exponential survival law.